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Gaiter and related methods

Abstract

An apparatus for being worn on a lower leg of a wearer. The apparatus includes a gaiter including one or more of the following: (a) a connector for connecting one or more accessories; (b) an adjustable strap including a retainer for connecting with a lace; (c) an adjustable stirrup; or (d) an upper end adapted for holding the gaiter in place on the lower leg.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present disclosure claims the benefit of Provisional Patent Application No. 63/299,696, filed on Jan. 14, 2022, the contents of which are incorporated by reference in it's entirety.

TECHNICAL FIELD

(1) This application relates to the clothing arts and, more particularly, to a gaiter and related methods.

BACKGROUND

(2) In many outdoor activities, it is desirable to wear a gaiter. Such gaiters typically run from the bottom of the foot to mid-calf and are made with sturdy weather-resistant materials. This design and fabrication protects the foot and lower leg from deep snow, wet underbrush or debris that you can pick up while hiking, cross-country skiing or mountaineering, as well as from insects and the elements.

(3) Many past designs of gaiters suffer from several limitations. For one, they can be difficult to put on, especially over footwear. For this reason, they are often made oversized, and thus do not stay up, especially when worn under extreme conditions, such as wilderness hiking or in inclement weather. Past gaiters also lack any means of holding or storing objects, which would be especially useful to provide ready access.

(4) Accordingly, a need is identified for a gaiter that addresses the foregoing issues and possibly others that have yet to be discovered.

SUMMARY

(5) According to a first aspect, an apparatus includes a gaiter including a connector for connecting one or more accessories.

(6) According to a second aspect, an apparatus includes a gaiter including an adjustable strap including a retainer for connecting with a lace of an item of footwear.

(7) According to a third aspect, an apparatus includes a gaiter including an adjustable stirrup.

(8) According to a fourth aspect, an apparatus includes a gaiter including an upper end having a drawstring.

(9) According to a fifth aspect, a method of using a gaiter in connection with footwear, the method includes adjusting the length of a loop associated with the gaiter.

(10) According to a sixth aspect, a method of using a gaiter includes attaching at least one accessory to a connector on the gaiter.

(11) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:
- (2) FIG. 1 schematically depicts a front perspective view of a gaiter, according to one or more embodiments shown and described herein;
- (3) FIG. 2 schematically depicts a rear perspective view of the gaiter of FIG. 1, according to one or more embodiments shown and described herein;
- (4) FIG. 3 schematically depicts a front perspective view of the gaiter of FIG. 1, according to one or more embodiments shown and described herein;
- (5) FIG. 3A schematically depicts a partial view of a drawstring and lock for the gaiter of FIG. 1, according to one or more embodiments shown and described herein;
- (6) FIG. 3B schematically depicts a partial view of a stirrup for the gaiter of FIG. 1, according to one or more embodiments shown and described herein;
- (7) FIG. 3C schematically depicts a partial view of a strap for attaching the gaiter of FIG. 1 to footwear, according to one or more embodiments shown and described herein;
- (8) FIG. 3D schematically depicts a partial view of the strap of FIG. 3C being attached to the gaiter, according to one or more embodiments shown and described herein;
- (9) FIG. 3E schematically depicts a partial view of a high friction surface provided on the inner surface of the gaiter of FIG. 1, according to one or more embodiments shown and described herein;
- (10) FIG. 4 schematically depicts a front perspective view of the gaiter of FIG. 1 with gear attached to the gaiter of FIG. 1, according to one or more embodiments shown and described herein; and
- (11) FIG. 5 depicts a flowchart of a method for using the gaiter of FIG. 1, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(12) In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration, specific embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the embodiments and like numerals represent like details in the various figures. Also, it is to be understood that other embodiments may be utilized, and that process or other changes may be made without departing from the scope of the disclosure. The following detailed description is not to be taken in a limiting sense, and the scope of the invention is defined only by the appended claims and their equivalents.

(13) Reference is now made to FIGS. 1 and 2, which depict a gaiter **10** according to one aspect of the disclosure. As can be understood, the gaiter **10** includes an elongated body **12** with a front flap **12a** designed to overlap with the remainder of the body and form a tubular sleeve. The elongated body **12** includes an upper end **13a** and an opposite lower end **13b**, where the tubular sleeves has openings at the upper end **13a** and the lower end **13b** for a user to extend a leg therethrough. The front flap **12a** may extend along the body **12** between the upper end **13a** and the lower end **13b**, and may connect to the body **12** by a releasable or removable connector, such as for example a hook and loop (i.e., VELCRO®) arrangement. For example, as depicted in FIGS. 3C and 3D, a first male connector (hook) in the form of an elongated strip **14a** may be provided on the body **12**, and a female connector (loop) also in the form of a strip **14b** provided along the corresponding interior surface of the flap **12a** for selectively coupling to the strip **14a**. However, it is contemplated and possible that the arrangement could be reversed, or other connectors such as snaps, magnets, or the like, could be used.

(14) Referring to FIGS. 1 and 3, the gaiter **10** may also include side flaps **12c**, **12d** that extend from the lower end **13b** of the body **12** to be configured for covering an upper sidewall of the footwear

corresponding to the wearer's ankle. The gaiter **10** may include a stirrup **20** that extends between the side flaps **12c**, **12d**, which is designed to loop around the wearer's foot and retain the lower end **13b** of the body **12** in contact or proximity with the wearer's shoe. Referring to FIGS. **3** and **3B**, the stirrup **20** may include a connector at each end adapted to connect with a similar, but oversized (at least in length) connector on the gaiter **10**, such as for example mating hook and loop fasteners **22**, **24**.

(15) A cover **26**, or flap, may also be provided for covering the connection once established, which may be along an inside surface of the side flaps **12c**, **12d** of the gaiter **10**. The cover **26** may also comprise a connector for connecting with the fasteners on the cover, the gaiter, or both. Again, this allows for the length of the stirrup **20** to be readily adjusted to accommodate different situations in terms of size of the corresponding midsection of the footwear.

(16) A retainer **28** may also be provided for retaining the stirrup **20** on the gaiter **10**, where the retainer **28** may be sewn or otherwise coupled to the respective side flap **12c**, **12d**, to allow for the stirrup **20** to extend therethrough. While FIG. **3B** depicts the fasteners **22**, **24** and the cover **26** attached to side flap **13a**, the gaiter **10** similarly includes fasteners **22**, **24** and the cover **26** on the side flap **13b**.

(17) Referring again to FIGS. **3C** and **3D**, adjacent to the portion of the body **12** including the strip **14a** is an adjustable strap **16** adapted to receive a retainer **18**, such as a clip, with a hook **18a** for engaging a shoe or boot lace **L**. The strap **16** may be coupled to the body **12** at one end of the strap **16** to extend from the body **12** and have a free end for attaching the body **12** to the wearer's footwear. The free end of the strap **16** may extend through an eyelet **18b** of the retainer **18** and then attach to the flap **12a** to retain the retainer **18** on the strap **16**. However, it is contemplated and possible that each end of the strap **16** is removably attachable to the body **12**.

(18) The strap **16** includes a connector **16a** adapted to connect with the strip **14a**, and thus may loop back to a suitable location in order to form a connection once the lace **L** is engaged. The flap **12a** may then be secured in place, engaging the connector **16a** with the strips **14a**, **14b**, thus protecting the strap **16** from becoming dislodged and releasing the retainer **18**. With both ends of the strap **16** attached to the body **12** with the retainer **18** coupled thereto, the retainer **18** may be engaged with the wearer's footwear by hooking the hook **18a** onto a lace or other feature on the footwear. The ability to adjust the length of the loop formed and thus the position of the retainer **18** allows for adjustments to be made to ensure a proper fit, which helps to ensure the gaiter remains in a proper position.

(19) Referring to FIGS. **3** and **3A**, along the upper end **13a** of the gaiter **10** that will lie adjacent to the wearer's knee, an adjustable retainer may also be provided, which may take the form of a drawstring **30** with a suitable lock **32**. The drawstring **30** may extend through a passage in the gaiter **10** to be coupled to the gaiter **10**, and thus be used to cinch and lock it at a particular diameter. Referring to FIG. **3E**, an opposite inner surface **12e** of the body **12** may also be provided with a tacky material or "grippy" surface that has a friction that is greater than a friction of rest of the body **12** for retaining the upper end **13a** of the body **12** in position. This feature, together with the drawstring **30**, helps to prevent the gaiter **10** from falling down along a lower portion of the leg of the wearer. The high friction surface may be in the form of a strip that is formed of a material with a friction that restricts the gaiter **10** from moving relative to the wearer, such as, for example, a rubber. Such a surface may also be provided along the lower open end of the gaiter **10**.

(20) FIG. **3** also illustrates that the body **12** of the gaiter **10** may comprise more than one material. For example, a lower portion of the body **12** for partially covering the foot and ankle may comprise a thicker, more durable/semi-rigid material, such as 1000D CORDURA® fabric. The upper portion may instead comprise a soft-shell, breathable fabric, such as smooth faced, stretchy, tightly-woven fabric that usually consists of nylon blended with elastane, in order to promote comfort.

(21) Turning back to FIGS. **3** and **4**, it can also be understood that various connectors **40** may be provided for connecting accessories or gear **50** to the gaiter **10**. The connectors may comprise a

plurality of so-called Modular Lightweight Load-carrying Equipment (MOLLE) loops **42**, which may be approximately 1.5" (3.8 cm) wide and located approximately 1" (2.5 cm) apart from each other. Multiple rows of such loops (four shown, but more or fewer may be provided) may be located along an outer side of the gaiter **10**, such as coupled to and extending along a sidewall **52** of the gaiter **10**, coupled to and extending along the front flap **12a**, or both, which thus allows for an object to be connected by being passed or woven through the loops. The loops **42** may also be provided along the front flap **12a** of the body **12**, and the drawstring **30** may pass through one of these loops on the front flap, such as the uppermost one. The connectors **40** may include a plurality of spaced rows of the loops **42** along both the front flap **12a** and the sidewall **52**.

(22) Using this arrangement, it can be appreciated that the gaiter **10** not only performs its protective function, but also serves as a location for holding or storing accessories, such as for example a knife in a sheath (see FIG. **4**, noting connected pouch **50**). As can be appreciated, this provides a removable storage system on the lower leg for survival gear, knives, tourniquets etc., so such accessories are readily accessible when needed, and without the need to remove root through a pocket or search a pack.

(23) In certain embodiments, the gear **50**) may be a protector **54** for attaching to the connectors **40** of one of the front flap **12a** and the sidewall **52** to provide protection for the wearer's legs. The protector **54** may be formed of a material that absorbs impact to the wearer's legs, such as when climbing over solid structures, the material being, for example, such as rubber. In embodiments, the protector **54** may be a 0.5 inch thick serrated strip pad that attaches to the connectors **40** along the front flap **12a** to protect a length of a shin of the wearer. In other embodiments, the protector **54** may be a shin-guard that is, for example, made of a rigid material such as plastic, metal, or the like, and attachable to one or more of the connectors **40**.

(24) Referring to FIG. **5**, a flowchart of a method **100** for using the gaiter **10** is depicted. The method **100** includes, at step **102**, adjusting the length of the stirrup **20** by moving the stirrup **20** relative to the loop fastener **22** and attaching the loop fastener **24** attached to the stirrup **20** to the loop fastener **22**, and covering the loop fasteners **22**, **24** with the cover **26**. The method **100** may further include adjusting a length of the strap **16** and retainer **18** by moving the strap **16** relative to the flap **12a** and attaching the connector **16a** to the connector **14a** on the body **12** of the gaiter **10**. The method **100** may further include adjusting a length of the drawstring **30** by moving the drawstring **30** through the lock **32** and releasing the lock **32** to lock the drawstring **30** in place. The method **100** includes, at step **104**, attaching an accessory, such as gear **50**, to one of the connectors **40** on the gaiter **10**.

(25) As used herein, the following terms have the following meanings:

(26) "A", "an", and "the" as used herein refers to both singular and plural referents unless the context clearly dictates otherwise. By way of example, "a compartment" refers to one or more than one compartment.

(27) "About," "substantially," or "approximately," as used herein referring to a measurable value, such as a parameter, an amount, a temporal duration, and the like, is meant to encompass variations of +/-20% or less, including +/-10% or less, +/-5% or less, +/-1% or less, and +/-0.1% or less of and from the specified value, in so far such variations are appropriate to perform in the disclosed invention. However, it is to be understood that the value to which the modifier "about" refers is itself also specifically disclosed.

(28) "Comprise", "comprising", and "comprises" and "comprised of" as used herein are synonymous with "include", "including", "includes" or "contain", "containing", "contains" and are inclusive or open-ended terms that specifies the presence of what follows e.g. component and do not exclude or preclude the presence of additional, non-recited components, features, element, members, steps, known in the art or disclosed therein.

(29) Although the invention has been described in conjunction with specific embodiments, many alternatives, modifications, and variations will be apparent to those skilled in the art. Accordingly,

it embraces all such alternatives, modifications, and variations that fall within the appended claims' spirit and scope. All publications, patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, the identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

Claims

1. An apparatus for partially covering a leg of a wearer including an item of footwear, comprising: a gaiter including an elongated body having a first flap with a closed position and an open position; and an adjustable strap including a first retainer adapted for connecting to the item of footwear, the adjustable strap being captured in place directly in-between the first flap and a portion of the body adapted for mating with an interior surface of the first flap when the first flap is in the closed position, and being removable from the gaiter when the first flap is in the open position.
2. The apparatus of claim 1, wherein the first flap is a front flap extending from an upper end to a lower end of the gaiter.
3. The apparatus of claim 1, wherein the first flap includes at least one loop adapted for connecting with an accessory.
4. The apparatus of claim 3, wherein the at least one loop comprises a plurality of vertically spaced loops extending along the first flap.
5. The apparatus of claim 1, wherein the retainer comprises an eyelet adapted for receiving the strap and a hook adapted for connecting with the item of footwear.
6. The apparatus of claim 1, wherein the gaiter includes a first connector and the first flap includes a second connector that connects with the first connector in the closed position to capture the adjustable strap in place.
7. The apparatus of claim 1, wherein the gaiter comprises an adjustable stirrup for engaging the item of footwear.
8. The apparatus of claim 7, wherein the gaiter comprises a second retainer forming a passage for receiving one end of the stirrup, the gaiter further including a second flap including a fastener for releasably attaching to the gaiter to secure the one end of the stirrup in place.
9. The apparatus of claim 1, wherein the gaiter comprises a plurality of loops positioned in rows on the elongated body.
10. The apparatus of claim 1, wherein the gaiter is further adapted to connect to a shin guard.
11. The apparatus of claim 1, further including a pouch connected to the gaiter.
12. An apparatus for partially covering a leg of a wearer including an item of footwear, comprising: a gaiter including an elongated body having a first flap adapted to move between a closed position and an open position; and an adjustable strap including a first retainer having an eyelet adapted for receiving the adjustable strap and including a hook for connecting to the item of footwear, the adjustable strap being captured in place directly in-between the first flap and a portion of the body adapted for mating with an interior surface of the first flap when the first flap is in the closed position, and being removable from the gaiter when the first flap is in the open position.
13. The apparatus of claim 12, wherein the first flap is a front flap extending from an upper end to a lower end of the gaiter.
14. The apparatus of claim 12, wherein the first flap includes at least one loop adapted for connecting an accessory to the gaiter.
15. The apparatus of claim 14, wherein the at least one loop comprises a plurality of vertically spaced loops extending along the first flap.
16. The apparatus of claim 12, wherein the gaiter includes a first connector and the first flap

includes a second connector for connecting with the first connector in the closed position to capture the adjustable strap in place.

17. The apparatus of claim 12, wherein the gaiter comprises an adjustable stirrup for engaging the item of footwear.

18. The apparatus of claim 17, wherein the gaiter comprises a second retainer for receiving one end of the stirrup, the gaiter further including a second flap including a fastener for releasably attaching to the gaiter to secure the one end of the stirrup in place.

19. The apparatus of claim 12, wherein the gaiter comprises a plurality of loops positioned in rows on the elongated body.

20. An apparatus for partially covering a leg of a wearer including an item of footwear, comprising: a gaiter including an elongated body including a front flap adapted to move between a closed position and an open position; and an adjustable strap including a retainer for engaging the item of footwear, the adjustable strap being captured in place directly in-between the front flap and a portion of the body adapted for mating with an interior surface of the front flap when the front flap is in the closed position, and being removable from the gaiter when the front flap is in the open position; an adjustable stirrup for engaging the item of footwear, wherein the gaiter comprises a side flap including a fastener for releasably attaching to the gaiter to secure the one end of the stirrup in place.
